

1 Empirical Analysis

We bring the cost-minimizing allocation model of Section ?? to a panel of 32 advanced economies over 1989–2022. The empirical strategy is a two-stage estimator: a first stage recovers each country’s structural sector size $\hat{\delta}_{ij}$, and a second stage uses these as interaction variables to estimate the slope and threshold behaviour of debt allocation.

1.1 Data and sector-size variables

Sources.

- **Debt holdings:** ?, sovereign debt by creditor sector.
- **Sector-size proxies:** World Bank, *Global Financial Development Database* (GFDD).
- **Panel:** 32 advanced economies, 1989–2022, for a total of $N = 780$ country–year observations. Coverage thickens after 2000; the full 32-country panel is available from 2007 onward.

Country–year proxies for structural sector size S_j .

$$\begin{aligned} s_{h,it} &= \frac{\text{Pension fund assets}_{it} + \text{Insurance assets}_{it}}{\text{GDP}_{it}} && \rightarrow S_h \text{ (non-bank capacity)} \\ s_{pb,it} &= \frac{\text{Bank deposit funding}_{it}}{\text{GDP}_{it}} && \rightarrow S_{pb} \text{ (private-bank size)} \\ s_{cb,it} &= \frac{\text{Central bank assets}_{it}}{\text{GDP}_{it}} && \rightarrow S_{cb} \text{ (central-bank size)} \end{aligned}$$

Foreign sector. No country-specific size proxy is available—foreign creditors are global balance sheets, not a local sector we can size in GFDD. We therefore run *no first-stage regression for foreigners*. Foreign debt is treated as the accounting residual

$$d_{f,it} = D_{it} - d_{h,it} - d_{cb,it} - d_{pb,it}. \quad (1)$$

1.2 Empirical strategy: two-stage estimator

Threshold. $\bar{D} = 100\%$ of GDP, chosen from the data: banking’s marginal absorption accelerates sharply once D_{it} crosses this level (discussed in the Motivation).

Stage 1—structural sector size $\hat{\delta}_{ij}$. For each country i and sector $j \in \{h, cb, pb\}$, regress sector share on total debt with country and year fixed effects, allowing a kink at $\bar{D}$:

$$s_{ijt} = \delta_{ij} + \delta_{jt} + \alpha_{1j}D_{it} + \alpha_{2j}\mathbf{1}\{D_{it} \geq \bar{D}\}(D_{it} - \bar{D}) + u_{ijt}. \quad (2)$$

We read $\hat{\delta}_{ij}$, the country fixed effect, as the part of country i ’s sector- j size that does not move with time, after stripping out variation explained by D_{it} and the kink.

Stage 2—debt allocation. The effective slope is a product of a baseline rate and a size scaling, with a kink at $\bar{D}$. In its original factored form,

$$d_{it}^j = \underbrace{\beta_j (1 + \gamma_j \hat{\delta}_{ij})}_{\text{baseline slope}} D_{it} + \underbrace{\mathbf{1}_{it} \eta_j (1 + \rho_j \hat{\delta}_{ij})}_{\text{post-threshold slope change}} (D_{it} - \bar{D}) + \varepsilon_{ijt}. \quad (3)$$

Expanding the two products gives the estimating equation reported in the tables:

$$d_{it}^j = \beta_j D_{it} + \theta_j (\hat{\delta}_{ij} \cdot D_{it}) + \mu_j \mathbf{1}_{it} (D_{it} - \bar{D}) + \nu_j \mathbf{1}_{it} (\hat{\delta}_{ij} \cdot (D_{it} - \bar{D})) + \varepsilon_{ijt}, \quad (4)$$

with $\theta_j = \beta_j \gamma_j$, $\mu_j = \eta_j$, and $\nu_j = \eta_j \rho_j$.

- β_j : baseline slope—sector j 's absorption per unit of D .
- θ_j : *size-scaling*—larger $\hat{\delta}_{ij}$ implies steeper absorption.
- μ_j, ν_j : post-threshold slope change and its size-scaling.

1.3 First-stage results: structural sector sizes

Table ?? reports estimates of (??).

Table 1 First stage: structural sector sizes

	(1) Non-bank s_{iht}	(2) Central bank s_{icbt}	(3) Private bank s_{ipbt}
α_{1j} : D_{it}	−0.203 (0.234)	0.016 (0.044)	−0.321 (0.265)
α_{2j} : $\text{highD} (D_{it} - \bar{D})$	−0.184 (0.203)	0.520*** (0.167)	0.856** (0.372)
N	547	718	732
Groups	31	32	32
R^2 (within)	0.436	0.603	0.199
Year / Country FE	Yes	Yes	Yes

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Takeaway.

- **Non-banks.** α_{1h}, α_{2h} are insignificant: non-bank size does not move with total debt, consistent with a *hard* capacity $\bar{\omega} S_h$.
- **Central- and private-bank size.** $\alpha_{2j} > 0$ and significant: banking-sector size *grows* once $D_{it} > \bar{D}$, consistent with banking's no-hard-ceiling property in the model.

1.4 Second-stage results: debt allocation

Table ?? reports estimates of (??).

Table 2 Second stage: debt allocation by sector

	Non-bank (d_h)	Central bank (d_{cb})	Private bank (d_{pb})
β_j : D_{it}	0.314*** (0.040)	0.053*** (0.012)	0.192*** (0.020)
θ_j : $\hat{\delta}_{ij} \cdot D_{it}$	0.0018*** (0.0005)	0.0131*** (0.0044)	−0.0002 (0.0007)
μ_j : threshold slope	−0.018 (0.171)	0.334** (0.153)	0.462** (0.220)
ν_j : threshold $\times \hat{\delta}$	−0.004 (0.003)	−0.055 (0.036)	−0.009 (0.008)
N	753	780	780
F	83.26	177.55	72.93
R^2	0.802	0.705	0.819

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Takeaways.

- **All sectors absorb part of new debt.** $\beta_j > 0$ across the three sectors.
- **Non-banks scale with size.** $\theta_h > 0$ and significant: countries with intrinsically larger non-bank sectors absorb a larger share of government debt at any given D . This maps to model prediction 2 (size scaling).
- **Banking takes over above $\bar{D}$.** $\mu_{cb} + \mu_{pb} \approx 0.80$: banking's marginal absorption jumps by roughly 80 percentage points once $D_{it} > 100\%$ of GDP. This maps to model prediction 4 (banking takeover).

Foreign debt residual. Using the fitted domestic components from (??), the implied foreign residual tracks the data: correlation = 0.779, slope = 0.744***, $R^2 = 0.608$, MAE = 8.79.

1.5 Concavity in debt absorption

Specification. For sector $j \in \{h, f\}$,

$$d_{it}^j = \beta_j D_{it} + \kappa_j D_{it}^2 + \theta_j (\hat{\delta}_h \cdot D_{it}) + \varepsilon_{it}, \quad (5)$$

where θ_j is included only for the non-bank sector. Pooled and country-FE estimates are reported separately, with and without $\hat{\delta}_h \cdot D$.

Table ?? reports estimates of (??).

Table 3 Concavity in debt absorption

	Non-bank (d_h)				Foreign (d_f)	
	Spec C: no $\hat{\delta}_h$		Spec B: with $\hat{\delta}_h \cdot D$		No $\hat{\delta}_h \cdot D$	
	Pooled	FE	Pooled	FE	Pooled	FE
$\beta(D)$	0.343***	0.402***	0.359***	0.299*	0.387***	0.392***
$\theta(\hat{\delta}_h \cdot D)$	—	—	0.0022***	0.0011*	—	—
$\kappa(D^2)$	−0.0003	−0.0016***	−0.0010	−0.0009	−0.0014***	−0.0009
N	780	780	701	701	747	747
R^2	0.735	0.381	0.801	0.381	0.773	0.454

Notes: Standard errors omitted for space. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Takeaways.

- **Non-banks.** Concavity ($\kappa_h < 0$) shows up *within country* but disappears once we control for size ($\hat{\delta}_h \cdot D$): the slowdown is driven by country-specific capacity S_h .
- **Foreigners.** Concavity ($\kappa_f < 0$) shows up *across countries* but not within: consistent with a *common* foreign ceiling $\bar{D}_f$.

Map to predictions. These two findings jointly support model prediction 3: the marginal absorption shares of non-banks and foreigners fall as D grows, but the source of concavity differs by sector—country-specific S_h for non-banks, a common $\bar{D}_f$ for foreigners.